Challenge 6: Better-Quasi-Ordering of Finite Graphs under Minor Relation

Definition 1 (Cumulative Hierarchy over Q)**.** Given a set Q, we denote by $\mathcal{P}^*$ the set of nonempty subsets of Q. We define for each ordinal α a set $V^*_\alpha(Q)$ recursively as follows:

- $V^*_0(Q) := Q$.
- $V^*_{\beta+1} := \mathcal{P}^*\mathcal{V}^*_\beta$
- $V^*_\lambda(Q) := \dot{\bigcup}_{\alpha<\lambda} V^*_\alpha(Q)$ for any limit ordinal λ.

The class given by the union of all these sets is denoted $V^*(Q)$.

Definition 2 (α-well-quasiordering)**.** Given a quasi-order $(Q, \leq)$, we define a relation $\leq^*$ on $V^*(Q)$ in terms of a 2-player game between Player I and Player II. A position in this game is a pair (X, Y) with X and Y in $V_*(Q)$. If X and Y are both in Q then this position is a win for Player II if $X \leq Y$ and a win for Player I otherwise. If not, then Player I names a set X', which must be X itself if $X \in Q$ and otherwise must be an element of X. Then Player II also names a set Y', which must be Y itself if $Y \in Q$ and otherwise must be an element of Y. After this, the new position is (X', Y'). We define $X \leq^* Y$ to mean that Player II has a winning strategy in this game with starting position (X, Y).

We say that $(Q, \leq)$ is *α-well-quasiordered* if the restriction of this game to $V^*_\alpha(Q)$ is well-quasiordered.

Note that $(Q, \leq)$ is better-quasi-ordered if and only if it is ω_1-well-quasi-ordered.

Conjecture 1.1. Let r be an ordinal number. If r is even, of the form $\alpha.2$, then finite planar graphs are α-well-quasi-ordered with respect to the minor relation. If r is odd, of the form $\alpha.2 + 1$, then finite graphs are α-well-quasi-ordered.